A Theory of Economic Slack

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Draft version: July 2026
Draft URL: pascalmichaillat.org/18/

CHAPTER 14.

Slackish Phillips curve

In chapter 13 we assumed that inflation was exogenously determined by price norms. However, the post-pandemic spike in inflation in the United States, which coincided with a very hot economy revealed that inflation might arise when the economy is inefficiently tight. This situation had not occurred in the United States in the modern, post-Volcker era of monetary policy since the US economy had not been inefficiently tight since the 1970s.

In this chapter, we insert a Phillips curve linking inflation to unemployment in the slackish business cycle model of chapter 13. The Phillips curve emerges from directed search: workers search for buyers offering the highest prices for their services, knowing of course that they will face more competition for high-paying jobs and less for low-paying jobs. The Phillips curve ensures divine coincidence: inflation is on target when the economy is at full employment. The Phillips curve is also convex in unemployment: inflation responds more strongly to unemployment when the economy is inefficiently tight than when the economy is inefficiently slack.

14.1. Overview of the model

The model follows the structure of the slackish business cycle model of chapter 13. Households purchase and consume services produced by other households. All the trades on the market for services are mediated by a matching function, which produces slack.

However, here the market is divided into many submarkets in which each household recruits workers to provide services for them. In each submarket, a household posts help-wanted ads that offer a specific price per service. Job search is directed: job seekers choose which submarket to join, depending on the price per service offered and of course the number of job seekers and help-wanted ads in that market. From a job seeker's perspective, the most attractive submarkets offer high prices, have many help-wanted ads, and have few other job seekers competing for the available positions.

Because search is directed, households must decide not only how many help-wanted ads they post but also what price they offer to pay for services rendered. A high price is expensive but allows households to recruit workers faster; a low price is cheap but imposes a long recruiting time.

This price-setting decision generates a Phillips curve. As usual in monetary models, to ensure that the Phillips curve is nondegenerate, we assume that households face price-adjustment costs—reflecting the type of fairness constraints discussed in chapter 5.

14.2. Slackish markets for services

Services are sold through long-term worker-household relationships. Once a worker has matched with a household, she becomes a full-time employee of the household, where she provides k services per unit time. She remains so until they separate, which occurs at rate λ.

To recruit workers, each household $j \in [0,1]$ posts V_j help-wanted ads. These ads specify the price P_j that the household plans to pay for the services purchased. Each help-wanted ad is serviced by κ recruiters per unit time. The recruiters are employees of household j devoted to recruiting other workers.

Workers of other households only work for or apply to one household. In that way, each household j constitutes a submarket. Each worker decides which submarket to join, based on the price P_j offered for services and the submarket tightness θ_j. The submarket tightness is the ratio of the number of help-wanted ads and job seekers: $\theta_j = V_j/U_j$, where U_j is the number of unemployed workers from other households vying for jobs in household j. The tightness matters because it determines how quickly workers find a job. So job seekers trade off how much they are paid on the job with how long it takes to find a job.

In each submarket j, a matching function $m(U_j, V_j)$ determines the flow of new matches based on the number of job seekers and help-wanted ads. The matching function m satisfies the assumptions from chapter 3. Tightness θ_j determines the job-finding and ad-filling rates, as usual.

14.3. Equilibrium aggregate supply

In each submarket j, there are l_{ij} employed workers from household i, and a total $l_j = \int_0^1 l_{ij}(t)\, di$ employed workers from all households. There are also U_{ij} unemployed workers from household i, and a total $U_j = \int_0^1 U_{ij}(t)\, di$ unemployed workers from all households. The total number of workers from household i who are attached to submarket j is $h_{ij} = l_{ij} + U_{ij}$, and the labor force attached to submarket j is $h_j = \int_0^1 h_{ij}(t)\, di$.

Because it takes time to find a job in any submarket j, not all h_j workers are employed. The unemployment rate in submarket j is

$$u_j = \frac{U_j}{h_j}.$$

We assume that labor flows in each individual submarket are balanced: the number of workers who find a job with household j at any point in time equals the number who leave their jobs. This assumption implies that the unemployment rate in submarket j is a function of submarket tightness: $u_j = u(\theta_j)$, where the function $u(\theta)$ is given by (8.8).

The equilibrium submarket supply is the amount of services provided at any point in time given the submarket tightness, and given that submarket flows are balanced:

$$y_j^s(\theta_j) = \left[1 - u(\theta_j)\right] kh_j. \tag{14.1}$$

The equilibrium aggregate supply is the amount of services provided at any point in time across the entire economy, given that all submarket flows are balanced: $y^s = \int_0^1 y_j^s(\theta_j)\, dj$.

14.4. Directed search and price-tightness tradeoff

Household j observes the prices and tightnesses in all submarkets, and picks the one that suits them best. Indeed, different household j may offer different prices P_j for services rendered. They may also post a different number of help-wanted ads and attract a different number of workers, generating a different tightness in submarket j—making it easier or harder to find a job there. In this section, we analyze how directed search links tightnesses and prices in all submarkets. This link will be the basis for the Phillips curve.

14.4.1. Arbitrage condition

When workers are unemployed, they do not receive any income. When they are employed, they produce k services per unit time, each paid a price P_j. Furthermore, in the submarket, the unemployment rate is $u(\theta_j)$, so a fraction $u(\theta_j)$ of the workers sent to submarket j are in unemployment, and a fraction $1 - u(\theta_j)$ are in employment. Accordingly, household i's expected income when they send h_{ij} workers to submarket j is $kP_j[1 - u(\theta_j)]h_{ij}$. The expected income per worker is $kP_j[1 - u(\theta_j)]$.

Household i does not prefer one employer or the other, so they pick their employer based solely on the expected income in the associated submarket j. They arbitrage across submarkets and send their workers to the submarket offering the highest expected income per worker. In their role as employer, households are aware of this arbitrage, so all households offer to pay a price for services that allows them to compete with other households. Arbitrage across submarkets therefore imposes that

$$P_j[1 - u(\theta_j)]$$

is the same across all submarkets j. Sellers direct their search toward the most attractive buyers, which induces competition across all buyers. Through competition, buyers set prices so sellers are indifferent across all buyers. If a buyer set a lower price than the rest, so the expected income on their submarket is lower than the rest, they would simply not attract any applicants.

By arbitrage, there is a price level p such that for all j,

$$P_j \cdot [1 - u(\theta_j)] = P \cdot [1 - u(\theta)], \tag{14.2}$$

where the aggregate tightness is the ratio of the aggregate number of help-wanted ads to the aggregate number of job seekers, given by

$$\theta = \frac{\sum_j V_j}{\sum_j U_j}.$$

14.4.2. Effect of price on tightness

The price posted by household j determines the tightness θ_j in their submarket, and therefore the ease with which the household is able to recruit workers. For instance, if the household offers to pay a high price for services, it attracts many workers—who want to work for a generous employer—so it can tap into a large pool of job seekers. This means that the household faces a low tightness, and that it can easily recruit workers.

This matters to household j because when it is harder to recruit—when help-wanted ads take longer to fill—they must devote more services to recruiting, and cannot consume as much. Indeed, the l_j workers in household j's employ produce $y_j = kl_j$ services. But the V_j help-wanted ads require each κ recruiters, so a total of $k\kappa V_j$ services are allocated to recruiting instead of consumption. Household j's consumption is therefore only

$$c_j = y_j - k\kappa V_j = k\left[l_j - \kappa V_j\right] = kh_j\left[1 - u_j - \kappa v_j\right], \tag{14.3}$$

where $u_j = U_j/h_j = 1 - l_j/h_j$ is the unemployment rate in submarket j and $v_j = V_j/h_j$ is the ad rate in submarket j.

Because we assume that flows are balanced in submarket j, we can express the gap between output and consumption of services using a matching wedge:

$$y_j = \left[1 + \tau(\theta_j)\right] c_j, \tag{14.4}$$

where the matching wedge $\tau(\theta_j)$ is given by (8.13).

Then, the arbitrage condition (14.2) tells us that when the price posted in submarket j is P_j, so the relative price in submarket j is $p_j = P_j/P$, then tightness in submarket j is

$$\theta_j(p_j) = u^{-1}\left(1 - \frac{1 - u(\theta)}{p_j}\right). \tag{14.5}$$

The function u^{-1} is strictly decreasing, so the submarket tightness $\theta_j(p_j)$ is decreasing in the relative price p_j. This means that a high relative price leads to low tightness and thus a low matching wedge $\tau(\theta_j)$. In fact, using again the results from appendix B, we see that the elasticity of the function $\theta_j(p_j)$ with respect to p_j is:

$$\epsilon_p^\theta = \frac{1}{\epsilon_\theta^u} \cdot \frac{-[1 - u(\theta)]/p_j}{1 - [1 - u(\theta)]/p_j} \cdot (-1) = \frac{-1}{[1 - \eta(\theta_j)]u(\theta_j)}, \tag{14.6}$$

where the elasticity ϵ_θ^u of $u(\theta)$ with respect to θ comes from (13.13) and is evaluated at θ_j, and we used $[1 - u(\theta)]/p_j = 1 - u(\theta_j)$.

In that way, employers have monopsonistic power in this economy. By offering higher prices for services, they attract more workers to their submarket, which allows them to recruit more easily. Conversely, if they choose to pay lower prices, fewer job seekers will join the submarket, and it will be difficult to recruit.

Household j therefore faces a tradeoff between the price per service, P_j, and the submarket tightness θ_j, which itself determines the amount of services that are devoted to matching, $\tau(\theta_j)$. All workers in household j are paid a price P_j per service. The expenditure by household j on workers therefore is

$$P_j y_j = \left[1 + \tau(\theta_j)\right] P_j c_j.$$

The effective price of services is not just P_j but $[1 + \tau(\theta_j)]P_j$. The price involves the price per service as well as the matching cost. Household j would prefer to pay a low price P_j and face a low tightness θ_j. But that is not possible, as (14.5) shows—hence the tradeoff.

The aim of the household is to minimize the effective price for services, $[1+\tau(\theta_j)]P_j$, which includes the cost of services consumed as well as the cost of services required for matching per unit of consumption. A low price P_j economizes on the price per service, but it imposes a high tightness θ_j and thus a high wedge $\tau(\theta_j)$.

14.4.3. Efficiency without price-adjustment costs

For reference, we describe what would happen in a submarket without any price-adjustment cost. Household j is free to set any price P_j she wants to minimize the effective price of services, $[1+\tau(\theta_j)]P_j$.

That is, the household chooses P_j to minimize $[1+\tau(\theta_j)]P_j$ subject to the arbitrage condition (14.2). Because of the arbitrage condition, the effective service price can be written

$$[1+\tau(\theta_j)]P_j = P \cdot [1-u(\theta)] \cdot \frac{1+\tau(\theta_j)}{1-u(\theta_j)}.$$

Hence, by choosing a price P_j, household j sets submarket tightness θ_j to minimize $[1+\tau(\theta_j)]/[1-u(\theta_j)]$, which is equivalent to maximizing $[1-u(\theta_j)]/[1+\tau(\theta_j)]$.

This in turn is equivalent to choosing tightness θ_j to maximize $1-u_j-\kappa v_j$. Indeed, notice by combining (14.1) and (14.4) that consumption by household j is

$$c_j = \frac{1-u(\theta_j)}{1+\tau(\theta_j)} k h_j.$$

But consumption is also given by (14.3), which shows that

$$\frac{1-u(\theta_j)}{1+\tau(\theta_j)} = 1-u_j-\kappa v_j.$$

This is equivalent to choosing the unemployment rate u_j to minimize $u_j+\kappa v(u_j)$, where the unemployment and ad rates are related by the Beveridge curve (8.5), which prevails here as flows are assumed to be balanced in submarket j. This is exactly the social planner's problem studied in chapter 9. Thus, tightness θ_j and unemployment rate u_j are chosen efficiently, so that condition (9.10) holds. Here we have just recovered the efficiency result of Moen (1997): directed search produces efficient market outcomes.

The key intuition is that directed search lets households internalize the planner's problem when choosing posted prices. In the absence of price-adjustment costs, each household's private pricing problem reproduces the planner's condition, so submarket tightness is efficient.

14.4.4. Introducing price rigidity

Generally, the unemployment rate is not efficient (chapter 12). To introduce inefficiency in the model, prices must be somewhat rigid.

Given that search is directed instead of random, prices are determined by buyers to minimize the effective cost of services—prices are not determined by norms here. To introduce price rigidity, we therefore assume that changing prices is costly for buyers. As in Rotemberg (1982), household j incurs a quadratic price-adjustment cost when inflation in submarket j departs from normal inflation π^n. The inflation in submarket j is

$$\pi_j(t) = \frac{\dot{P}_j(t)}{P_j(t)}. \tag{14.7}$$

The flow disutility caused by actual inflation deviating from normal inflation is

$$\frac{\varpi}{2} \cdot \left(\pi_j - \pi^n\right)^2.$$

The parameter $\varpi > 0$ governs the price-adjustment cost. This quadratic cost appears in the household j's utility function.

What does this price-adjustment mean? What does it capture in reality? First of all, when prices fall, or increase less than normal, workers in household j feel shortchanged. Indeed, the price p_j directly determines their salary, $p_j k$. And we saw in chapter 6 that workers' morale dips when their wages do not grow as expected, and that their performance may dip as well. The quadratic cost reflects the cost of restoring workers' morale, or the cost caused by diminished performance, when price growth is below the normal growth π^n.[1]

When prices rise, or increase more than normal, household j might be unhappy to pay more for the same services. We saw in chapter 6 that higher-than-normal inflation upsets customers, who feel unfairly treated when they shop and have to pay more for the same good or service. Here we assume that this displeasure with higher prices takes the form of a quadratic cost when price growth is above the normal growth π^n.

[1]The typical Keynesian wage-floor idea is that workers do not tolerate a drop in the level of nominal wages. But as Okun (1981, p. 93) notes, the same idea might apply to the growth rate of nominal wages, as we assume here: "Workers are likely to develop some notion of how fast they expect their wages to rise.... Any subsequent slowdown below the norm must be justified and explained to experienced workers. If the employer stresses the norm, a Keynesian wage-floor phenomenon may apply to the rate of increase of wages, and not just to their level."

14.5. Saving and pricing by households

We now present and solve the optimization problem of households. Households have essentially two decisions to make: how much income to save, and how to set prices for the services they purchase. From the optimal saving decision we will derive the aggregate demand curve, and from the optimal pricing decision the Phillips curve. We will then use these curves together with the aggregate supply curve to solve the model.

14.5.1. Household's utility function

Just as in chapter 13, household j cares about their consumption of services and their social status, derived from their relative real wealth. Here too, the household's nominal wealth $B_j(t)$ is held in government bonds. The household's relative nominal wealth is $B_j(t) - B(t)$, where $B(t) = \int_0^1 B_j(t)\,dj$ is average nominal wealth in the economy. The household's relative nominal wealth is $[B_j(t) - B(t)]/P(t)$, where the price level $P(t)$ is given by (14.2). In addition households incur disutility from changing the prices they pay for services.

Hence, household j maximizes the discounted sum of flow utilities,

$$\int_0^\infty \exp(-\delta t)\left[c_j(t)^{1-\alpha} + \frac{1}{a}\cdot\frac{B_j(t) - B(t)}{P(t)} - \frac{\varpi}{2a}\cdot\left(\pi_j(t) - \pi^n\right)^2\right] dt, \tag{14.8}$$

where the parameter $\delta > 0$ is the time discount rate; the parameter $a > 0$ indicates the taste for consumption relative to other components of the utility function—saving and changing prices; and the parameter ϖ governs the cost of changing price.

14.5.2. Household's budget constraint

Household j's nominal budget constraint is similar to that in chapter 13 except that the income of the household now comes from employment in different households that might pay different prices for services. The budget constraint therefore is

$$\dot{B}_j(t) = i(t)\cdot B_j(t) + \int_0^1 P_i(t) y_{ij}(t)\,di - P_j(t) y_j(t) - T(t),$$

where $T(t)$ is a lump-sum tax levied by the government.

Given that the unemployment rate is $u(\theta_i)$ in each submarket i, and given the arbitrage condition (14.2), the household's income can be rewritten as follows:

$$\begin{aligned}\int_0^1 P_i(t) y_{ij}(t)\, di &= \int_0^1 P_i(t)[1 - u(\theta_i(t))]kh_{ij}(t)\, di \\ &= p(t)[1 - u(\theta(t))]k \int_0^1 h_{ij}(t)\, di \\ &= p(t)[1 - u(\theta(t))]kh_j.\end{aligned}$$

Given the matching wedge (14.3) and tightness-price link (14.5), the household's expenditure can be written as a function of consumption and price:

$$P_j(t) y_j(t) = P_j(t) \left[1 + \tau\left(\theta_j\left(\frac{P_j(t)}{P(t)}\right)\right)\right] c_j(t).$$

Accordingly, the nominal budget constraint can be written as

$$\dot{B}_j(t) = i(t) \cdot B_j(t) + p(t)[1 - u(\theta(t))]kh_j - P_j(t) \left[1 + \tau\left(\theta_j\left(\frac{P_j(t)}{P(t)}\right)\right)\right] c_j(t) - T(t).$$

Then, just as in chapter 13, we translate the nominal budget constraint into a real budget constraint:

$$\dot{b}_j(t) = r(t) \cdot b_j(t) + [1 - u(\theta(t))]kh_j - p_j(t) \left[1 + \tau\left(\theta_j\left(p_j(t)\right)\right)\right] c_j(t) - \frac{T(t)}{P(t)}, \tag{14.9}$$

where $b_j(t) = B_j(t)/P(t)$ is real bond holdings. The real interest rate is defined by $r(t) = i(t) - \pi(t)$, where the inflation rate is $\pi(t) = \dot{P}(t)/P(t)$.

14.5.3. Law of motion for household's price

Besides the budget constraint, household j faces another constraint: the law of motion for the price it pays for services. The law of motion follows from (14.7), and says that any price change requires some inflation, which itself is costly. The law of motion for the nominal price $P_j(t)$ is

$$\dot{P}_j(t) = \pi_j(t) P_j(t).$$

Hence, using a result of the sort of (13.5), we find that the law of motion for the relative price $p_j(t) = P_j(t)/P(t)$ is

$$\dot{p}_j(t) = \left[\pi_j(t) - \pi(t)\right] p_j(t). \tag{14.10}$$

14.5.4. Household's problem

We now solve household j's maximization problem, which is to maximize (14.8) subject to the laws of motion (14.9) and (14.10). We do so using a Hamiltonian (result A.25):

$$\begin{aligned}\mathcal{H}(t, c_j(t), \pi_j(t), b_j(t), p_j(t)) &= c_j(t)^{1-\alpha} + \frac{1}{a}\left[b_j(t) - b(t)\right] - \frac{\varpi}{2a}\left[\pi_j(t) - \pi^n\right]^2 \\ &+ \psi_j^b(t)\left\{r(t)b_j(t) + [1 - u(\theta(t))]kh_j - \left[1 + \tau\Big(\theta_j\big(p_j(t)\big)\Big)\right]p_j(t)c_j(t) - \frac{T(t)}{P(t)}\right\} \\ &+ \psi_j^p(t)\left[\pi_j(t) - \pi(t)\right]p_j(t).\end{aligned}$$

The control variables are consumption $c_j(t)$ and inflation $\pi_j(t)$. The state variables are real bond holdings $b_j(t)$ and relative price $p_j(t)$. The costate variables are $\psi_j^b(t)$, which applies to the law of motion of real bond holdings (14.9), and $\psi_j^p(t)$, which applies to the law of motion of the relative price (14.10).

We now study the first-order conditions that characterize the household's optimal behavior. We begin with the first-order condition with respect to consumption: $d\mathcal{H}/dc_j = 0$. It gives

$$(1 - \alpha)c_j(t)^{-\alpha} = \psi_j^b(t)\left[1 + \tau(\theta_j(t))\right]p_j(t). \tag{14.11}$$

Next we turn to the first-order condition with respect to inflation: $d\mathcal{H}/d\pi_j = 0$. It yields

$$\frac{\varpi}{a}\left[\pi_j(t) - \pi^n\right] = \psi_j^p(t)p_j(t). \tag{14.12}$$

The next first-order condition is that with respect to real bond holdings: $d\mathcal{H}/db_j = \delta\psi_j^b(t) - \dot{\psi}_j^b(t)$. It gives the same condition as in chapter 13:

$$\dot{\psi}_j^b(t) = [\delta - r(t)]\,\psi_j^b(t) - \frac{1}{a}. \tag{14.13}$$

The final first-order condition is that with respect to the relative price: $d\mathcal{H}/dp_j = \delta\psi_j^p(t) - \dot{\psi}_j^p(t)$. This condition becomes

$$\begin{aligned}\dot{\psi}_j^p(t) - \delta\psi_j^p(t) &= \psi_j^p(t)[\pi(t) - \pi_j(t)] \\ &+ \psi_j^b(t)c_j(t)[1 + \tau(\theta_j(p_j(t)))]\left[1 + p_j(t)\frac{\tau'(\theta_j(t))}{1 + \tau(\theta_j(p_j(t)))}\theta_j'(p_j(t))\right].\end{aligned}$$

We can rewrite the following term:

$$p_j(t)\frac{\tau'(\theta_j(t))}{1+\tau(\theta_j(p_j(t)))}\theta_j'(p_j(t)) = \frac{\theta_j(t)}{1+\tau(\theta_j(p_j(t)))}\cdot\tau'(\theta_j(t))\cdot\frac{p_j(t)}{\theta_j(t)}\cdot\theta_j'(p_j(t)) = \epsilon_\theta^{1+\tau}\cdot\epsilon_p^\theta,$$

where the elasticity $\epsilon_\theta^{1+\tau}$ is given by (6.12) and the elasticity ϵ_p^θ is given by (14.6), both evaluated at $\theta_j(t)$. Hence, the term boils down to

$$p_j(t)\frac{\tau'(\theta_j(t))}{1+\tau(\theta_j(p_j(t)))}\theta_j'(p_j(t)) = -\frac{\eta(\theta_j(t))}{1-\eta(\theta_j(t))}\cdot\frac{\tau(\theta_j(t))}{u(\theta_j(t))}.$$

Noting lastly that $[1+\tau(\theta_j(t))]c_j(t) = y_j(t)$, we simplify the first-order condition to

$$\dot{\psi}_j^p(t) = [\delta+\pi(t)-\pi_j(t)]\psi_j^p(t) + \psi_j^b(t)y_j(t)\left[1-\frac{\eta(\theta_j(t))}{1-\eta(\theta_j(t))}\cdot\frac{\tau(\theta_j(t))}{u(\theta_j(t))}\right]. \tag{14.14}$$

14.5.5. Symmetric solution

We focus on a symmetric solution to the model, in which all households behave the same, and all submarkets are therefore the same. In this symmetric situation, all submarket tightnesses θ_j are the same and equal the aggregate tightness θ. All submarket prices P_j are the same and equal the price level P, so all relative prices p_j equal 1, and submarket inflation π_j is the same as aggregate inflation π.

We can drop the index j on all individual variables.

The first-order conditions are necessary for an interior solution to the household's problem. So the symmetric solution must satisfy them. Under symmetry, the conditions (14.11), (14.12), (14.13), and (14.14) simplify as follows:

$$(1-\alpha)c(t)^{-\alpha} = \psi^b(t)\left[1+\tau(\theta(t))\right] \tag{14.15}$$

$$\frac{\varpi}{a}\left[\pi(t)-\pi^n\right] = \psi^p(t) \tag{14.16}$$

$$\dot{\psi}^b(t) = [\delta-r(t)]\,\psi^b(t) - \frac{1}{a} \tag{14.17}$$

$$\dot{\psi}^p(t) = \delta\psi^p(t) + \psi^b(t)y(t)\left[1-\frac{\eta(\theta(t))}{1-\eta(\theta(t))}\cdot\frac{\tau(\theta(t))}{u(\theta(t))}\right]. \tag{14.18}$$

14.6. Monetary policy

Before solving the model, we must specify monetary policy. Monetary policy has two dimensions in the model: setting interest rates and communicating around the inflation that can be expected in the future.

First, the central bank maintains a fixed real interest rate r. This is achieved by setting the nominal interest rate to $i(t) = r + \pi(t)$ at any point. There is no Taylor rule here: the central bank adjusts the nominal interest rate one-for-one with inflation to keep the real interest rate fixed. The central bank focuses on the real interest rate because this is the relevant price in the model: it is the relevant interest rate for aggregate demand.

Second, we assume that through effective communication and management of expectations, the Federal Reserve anchors people's inflation expectations to the inflation target, which is 2% in practice (FOMC 2026). Hence, we assume that normal inflation is just the Federal Reserve's inflation target: $\pi^n = \pi^*$. That is, the Federal Reserve manages to convince people that prices will grow at their target inflation rate. This assumption plays a central role in producing the divine coincidence, but it seems quite realistic. In the United States, since 1995, the long-run forecast of inflation elicited in the Survey of Professional Forecasters has been exceedingly stable, ever so slightly above 2% (Hazell et al. 2022, figure 2).

14.7. Equilibrium aggregate demand

We now derive the aggregate demand in the model. Just as in chapter 13, the aggregate demand is determined by two first-order conditions: (14.15) and (14.17).

These equations are the same equations as in chapter 13, so the aggregate-demand block is not affected by directed search or endogenous inflation. The reason is that by setting a real interest rate, monetary policy isolates aggregate demand from inflation. Inflation matters for aggregate demand only insofar as it affects the real interest rate. With the typical Taylor rule assumed in New Keynesian model, inflation and real rate are linked. But that assumption appears there for convenience—to ensure determinacy of the equilibrium (Cochrane 2011). Here the equilibrium is determinate without it, so we simplify the analysis by making the reasonable assumption that the central bank fixes directly the price that matters: the real interest rate.

As in chapter 13, to respect the transversality condition and no-Ponzi condition, the costate variable ψ^b cannot diverge to ∞, and because the differential equation (14.17) is a source, the costate variable must jump to its equilibrium value at time 0, which is again given by equation (13.9): $\psi^b = 1/[(\delta - r)a]$.

Then, again as in chapter 13, the household's consumption is a function of the real interest rate and aggregate tightness. The function is obtained by plugging the value of the costate variable in (14.15). This gives again equation (13.10), from which we derive the same aggregate demand as in equation (13.11):

$$y^d(\theta, r) = \frac{[(1-\alpha)(\delta - r)a]^{1/\alpha}}{[1+\tau(\theta)]^{1/\alpha - 1}}. \tag{14.19}$$

14.8. Phillips curve

Finally, we derive the Phillips curve in the model. The Phillips curve comes from the two first-order conditions that we have not used so far: (14.16) and (14.18).

14.8.1. Expression of the Phillips curve

Equation (14.16) simply says that inflation is a linear function of the costate variable associated with the relative price: $\pi(t) = \pi^* + a\psi^p(t)/\varpi$.

The costate variable $\psi^p(t)$ is given by (14.18), which is a linear first-order differential equation. The equation is a source since $\delta > 0$. So if the costate variable does not jump to the equilibrium point of the differential equation, it exhibits explosive dynamics: going to ∞ as $t \to \infty$. Such dynamics imply that inflation would also diverge to $\pm\infty$, which cannot be optimal as that generates infinitely negative utility. So this costate variable must jump to the equilibrium point, just like the costate variable associated with real wealth:

$$\psi^p = \frac{y(\theta)}{\delta(\delta - r)a} \cdot \left[\frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right].$$

Accordingly, inflation jumps at time 0 to the following value:

$$\pi = \pi^* + \frac{y(\theta)}{\varpi\delta(\delta - r)} \cdot \left[\frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right]. \tag{14.20}$$

Equation (14.20) is the Phillips curve of the model. It links inflation to aggregate tightness, which is itself determined from the aggregate demand and supply curves. It has a few striking properties, which we review now.

14.8.2. Properties of the Phillips curve

The first property of the Phillips curve is that the divine coincidence holds: inflation is on target ($\pi = \pi^*$) whenever the economy operates efficiently—whenever the economy is at full employment ($u = u^*$). This can be seen because, when the economy operates efficiently, the term in square brackets in the Phillips curve is 0, as we established in equation (9.11).

In fact the term in square brackets in (14.20) measures the productive inefficiency of the economy. When the economy is inefficiently tight, the term is positive, so inflation is above target. When the economy is inefficiently slack, the term is negative, so inflation is below target. Furthermore, the response of inflation to departures from efficiency is governed by the price-adjustment cost ϖ. When prices are very costly to change, ϖ is large and inflation does not respond much to changes in tightness. By contrast, when prices are easy to change, ϖ is small and inflation responds much more forcefully to changes in tightness.

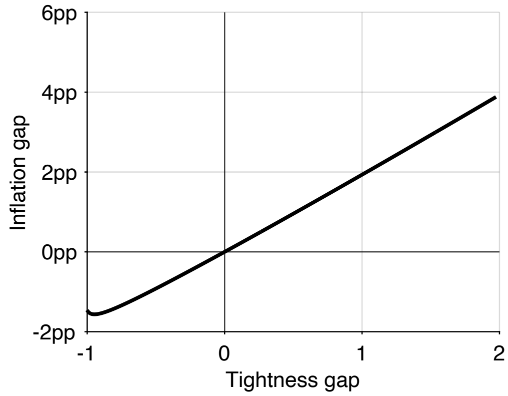
Thanks to the divine coincidence, the central bank does not face a tradeoff between inflation and unemployment: by bringing the unemployment rate to the FERU, it mechanically brings inflation to its target. Of course, the divine coincidence arises because we assumed that people take the inflation target as normal inflation. If that's not the case, the divine coincidence breaks down, as we discuss below. But when it's the case, the government automatically achieves its price mandate when it achieves its employment mandate.

We saw that inflation is above target when tightness is inefficiently high and below target when tightness is inefficiently low. But, more generally, inflation is strictly increasing in tightness around full employment and above. This can be seen because output $y(\theta)$, matching elasticity $\eta(\theta)$, matching wedge $\tau(\theta)$ are increasing in tightness, while the unemployment rate $u(\theta)$ is strictly decreasing in tightness. So above full employment ($\theta > \theta^*$), the term in square brackets is positive and growing with tightness and output is positive and growing with tightness so inflation is growing with tightness. Around full employment, the term in square brackets is 0, so the slope of the Phillips curve is determined by the derivative of the term in square brackets, which is strictly positive. Once again, inflation is increasing in tightness.

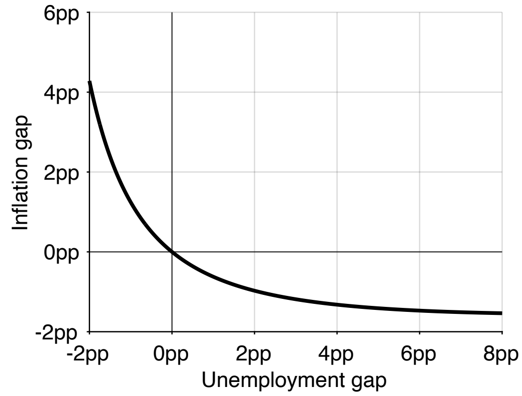
The logic actually continues to hold even below full employment ($\theta < \theta^*$) as long as output is large enough. Below full employment there are two opposing forces: output is increasing with tightness; the term in square brackets is increasing with tightness, but that term is negative and becoming less negative. Hence the product of the two terms is negative and could be increasing or decreasing. When tightness is close to full employment and the term in square brackets is close to 0, the change in the term dominates, so that inflation is increasing in tightness. When tightness is much lower and the term is square brackets is very negative, the negative change in output dominates, and inflation might decrease with tightness.

To illustrate the properties of the Phillips curve, we plot it in the calibrated model. Just as in chapter 13, we use the calibration from table 10.2, with in addition labor productivity normalized to $k = 1$, the discount rate set to $\delta = 91\%$, and the real interest rate set to $r = 0.5\%$. Under this calibration, the efficient tightness is $\theta = 1.03$ and the efficient slack rate is $u^* = 4.5\%$.

Here there is an important additional parameter to calibrate: the price-adjustment cost ϖ, which determines the slope of the Phillips curve. We calibrate it from microevidence collected by Zbaracki et al. (2004). Using time-and-motion methods, they study the pricing process of a large industrial firm. They find that the physical, managerial, and customer costs of changing prices amount to 1.22% of the firm's revenue in a given year. That year,



A. Tightness-inflation curve



B. Unemployment-inflation curve

FIGURE 14.1. Phillips curves in the calibrated model

The Phillips curve is given by equation (14.20). Tightness and unemployment are related by equation (8.8). The model is calibrated as in table 10.2, with in addition labor productivity normalized to $k = 1$, the discount rate set to $\delta = 91\%$, the real interest rate set to $r = 0.5\%$, and the price-adjustment costs set to $\varpi = 58$.

the firm changed the price of 25% of its products, and most prices changed by about 4%. Based on this evidence, we calibrate ϖ so that if inflation is 4% on 25% of the services purchased by a household, the cost amounts to 1.2% of the household's purchases (which correspond to the revenue of the household's workers) in utility terms. So we set the parameter so that

$$\frac{\varpi}{2a} \times 0.04^2 \times 0.25 = 0.0122 \times y,$$

where a is the taste for consumption and y is the real expenditure or income in the model. At the average tightness of 0.65, $y = 0.943$ and $a = 1$, so $\varpi = 58$.

Under this calibration, the Phillips curve is plotted in figure 14.1A. The non-monotonicity of the Phillips curve is observable but not relevant in practive. Inflation is increasing in tightness over the relevant range of tightness: for any tightness above 0.08. This means that generally, as expected, inflation rises as the economy becomes tighter. We also see that the Phillips curve grows broadly linear in tightness over a wide range: for tightness between 0.08 and 3.

14.8.3. The unemployment-inflation Phillips curve

The tightness-inflation Phillips curve appears almost linear under the calibration considered (figure 14.1A). However, once we represent the Phillips curve in a more traditional unemployment-inflation plane, the curvature of the Phillips curve changes drastically (figure 14.1B). The unemployment rate is a strictly decreasing function of tightness, given by (8.8), so tightness is a strictly decreasing function of the unemployment rate, $\theta(u)$. The figure plots $\pi(\theta(u))$, where $\pi(\theta)$ is given by (14.20).

As expected, the inflation rate is decreasing in the unemployment rate, but interestingly, inflation is now a convex function of unemployment. In the calibration, the point of divine coincidence occurs when inflation is $\pi^* = 2\%$ and unemployment is $u^* = 4.5\%$. The Phillips curve goes through it, and it is much steeper when the unemployment rate is inefficiently low than when it is inefficiently high. When the unemployment gap goes up to 8pp, the inflation gap falls by less just 1.5pp—so inflation falls to just 0.5%. But when the unemployment gap drops to -2pp, the inflation gap increases to 4.2pp, so inflation climbs to 6.2%.[2]

Although inflation varies smoothly with unemployment, the model produces relatively large fluctuations in inflation when the economy is inefficiently tight and relatively small fluctuations in inflation when the economy is inefficiently slack. Conversely, the model produces relatively small fluctuations in unemployment when the economy is inefficiently tight and relatively large fluctuations in unemployment when the economy is inefficiently slack. Here we obtain the result numerically; in appendix H, we establish analytically that the Phillips curve $\pi(u)$ is strictly convex under the common assumption that the matching function is Cobb-Douglas and symmetric: $m(u, v) = \mu\sqrt{uv}$.

14.8.4. Nonlinearity of the model

An interesting numerical property of the Phillips curve is that it is almost linear in tightness under this calibration (figure 14.1A). But that does not mean that the model is linear, because the aggregate supply curve is sharply nonlinear (figure 8.1B). For instance, aggregate demand shocks have small effects on tightness in a slack economy—they affect mostly output—but large effects on tightness in a tight economy. As inflation is linear in tightness, it behaves in a similar fashion: under aggregate demand shocks, it responds little when the economy is slack and much more when the economy is tight.

Relatedly, one wonders how the Phillips curve can have such different curvature depending on how it is represented: in a tightness-inflation plane or an unemployment-inflation plane. In fact, the near-linearity in the tightness-inflation representation is consistent with the strong curvature in the unemployment-inflation representation because unemployment is a nonlinear function of tightness through the Beveridge mapping $u(\theta)$, so a nearly linear $\pi(\theta)$ can generate a convex $\pi(u)$.

To conclude this discussion of nonlinearity, notice that in an output-unemployment plane, the aggregate supply curve is linear, given by $y^s(u) = (1 - u)kh$. So unemployment responds in a similar fashion to aggregate demand shocks whether the economy is slack or tight. However, when the economy is tight, inflation responds sharply to unemployment because of the convex Phillips curve, so it also responds sharply to aggregate

[2]The shape of the Phillips curve simulated in figure 14.1B bears an uncanny similarity with the shape of the original curve estimated by Phillips (1958, figure 1) from UK data for the years 1861–1913.

demand shocks. By contrast, when the economy is slack, inflation barely responds to unemployment, so it responds little to aggregate demand shocks.

Of course, whether we analyze the model through tightness or unemployment, the properties remain the same, although in the former case the nonlinearity comes from the aggregate supply curve while in the latter case it comes from the Phillips curve.

14.8.5. Intuition behind the Phillips curve

We have seen that price dynamics are driven by the amount of slack in the economy. When the economy is inefficiently slack, buyers are able to lower the prices they pay below what is normal. Conversely, when the economy is inefficiently tight, buyers are pushed to accept higher prices than what is normal. What is the intuition for this behavior?

When inflation is above normal, a buyer can reduce its price-adjustment cost by lowering its rate of inflation. Since pricing is optimal, however, there cannot exist any profitable deviation from the current situation. This means that the buyer must incur a commensurate cost when it lowers its rate of inflation. With lower inflation, the price offered by the buyer drops relative to the prices of other buyers. The absence of profitable deviation imposes that the price reduction is costly, so the price must already be below the effective-price-minimizing level. And since submarket tightness is inversely related to the buyer's price, submarket tightness must be above the effective-price-minimizing tightness, which is just the efficient tightness (section 14.4.3). In other words, when inflation is above normal, tightness must be inefficiently high—otherwise profitable deviations would exist for price-setting households.

The same logic holds when inflation is below normal. Then a buyer can reduce its price-adjustment cost by raising its rate of inflation. With higher inflation, the price offered by the buyer rises relative to the prices of other sellers. The absence of profitable deviation imposes that the price increase is costly, so the price must already be above the effective-price-minimizing level. And since submarket tightness is inversely related to the buyer's price, submarket tightness must be below the effective-price-minimizing level, which is the efficient tightness. So when inflation is below normal, tightness must be inefficiently low.

In this chapter, we have assumed that buyers set prices: they raise them when few wokers are available to provide services and lower them when many workers are available. However, the results remain exactly the same if it is sellers and not buyers who set prices, as Michaillat and Saez (2024) show. In that case, workers are able to raise prices for their services when they are in high demand, and are forced to lower their prices when demand slumps.

14.9. Solution of the model in normal times

We now solve the slackish business cycle model. We concentrate on normal times, when the nominal interest rate is positive. After that we will turn to the zero lower bound, when the nominal interest rate is at zero.

By symmetry, the aggregate supply is simply:

$$y^s(\theta) = [1 - u(\theta)]\, kh. \tag{14.21}$$

The aggregate demand is given by (14.19). The tightness that solves the model equalizes aggregate supply and demand:

$$y^d(\theta, r) = y^s(\theta). \tag{14.22}$$

Aggregate demand, aggregate supply, and solution are illustrated in figure 13.1B.

Then the inflation rate is given by the Phillips curve (14.20) and the unemployment rate by (8.8), and so on.

14.10. Implications for the conduct of monetary policy

This section now reviews a few implications of the slackish Phillips curve for the conduct of monetary policy.

14.10.1. Satisfying the dual mandate

In the United States, the Federal Reserve has a dual mandate: maintaining the economy at full employment while keeping prices stable. In the model, both mandates can be achieved through monetary policy. By selecting the real interest rate r, monetary policy moves the aggregate demand curve (14.19) along the aggregate supply curve. To reach the dual mandata, policy must bring the aggregate demand curve to the point where tightness is efficient ($\theta = \theta^*$). At this point, it is guaranteed through the Beveridge curve that the economy is at full employment ($u = u^*$) and along the Phillips curve that prices are stable ($\pi = \pi^*$).

We refer to the real interest rate that brings the economy to full employment as the full-employment rate of interest (FERI), and denote it by r^*. In the model, because of the divine coincidence, the FERI also ensures that prices are stable. The FERI is defined as the real interest rate that guarantees that aggregate demand and supply curve intersect at the efficient tightness:

$$y^d(\theta^*, r^*) = y^s(\theta^*).$$

From the expressions (14.19) and (14.21) for the aggregate demand and supply curves, we obtain an expression for the FERI:

$$r^* = \delta - \frac{1}{a} \cdot \frac{1 + \tau(\theta^*)}{(1 - \alpha) c(\theta^*)^{-\alpha}}. \tag{14.23}$$

The corresponding nominal interest rate is just $i^* = r^* + \pi^*$.

The Fed can achieve its dual mandate only if $i^* \geq 0$. If $i^* < 0$, then the dual-mandate solution would violate the zero lower bound constraint that $i \geq 0$. In that case the central bank would resort to setting $i = 0$. The zero lower bound become binding when $r^* < -\pi^*$. We see from (14.23) that this situation arises the taste for consumption a is low enough, so that aggregate demand is depressed enough.

At the zero lower bound, monetary policy can no longer pin down the real interest rate independently of inflation: because the nominal interest rate is zero, the real interest rate is just negative inflation. So the normal-times decomposition of the model between the aggregate-demand-aggregate-supply block and the Phillips-curve block no longer applies. Below we tackle the analysis of the zero lower bound.

14.10.2. Choice of target variables

In the model, since the full-employment and price-stability mandates of the Fed coincide, aiming for the inflation target or the efficient unemployment rate is completely equivalent. In practice, however, the Fed should target the variable that is the most volatile—to observe more clearly departures from the dual mandate.

The slope of the Phillips curve in turn determines which of inflation and unemployment is the most volatile. If the Phillips curve is steep, aggregate-demand shocks will mostly generate movements in inflation, and targeting inflation will be easier. By contrast, if the Phillips curve is flat, aggregate-demand shocks will mostly generate movements in unemployment, and targeting unemployment will be easier.

In the model, the Phillips curve appears convex: flat when unemployment is inefficiently high, and steep when unemployment is inefficiently low. The implication is that the Fed should target the FERU u^* when the economy is inefficiently slack; and it should target an inflation rate of 2% when the economy is inefficiently tight.

We saw that the inflation-tightness Phillips curve was broadly linear, which means that inflation and tightness vary commensurately. Hence, there is a corrolary of the results for situations in which it is easier to measure tightness than inflation. In those cases, the Fed should target the FERU u^* when the economy is inefficiently slack and the efficient tightness θ^* when the economy is inefficiently tight. Shifting between unemployment and tightness in slumps and booms would give the Fed the maximum amount of information and precision to conduct policy.

14.11. Ineffective monetary communication

If monetary communication is ineffective, the divine coincidence breaks down. What we mean by ineffective monetary communication is that the inflation that people consider normal, π^n, is not aligned with the inflation target, π^*.

With ineffective monetary communication, the derivations are the same, except that π^* is replaced by π^n in the Phillips curve (14.20). It is as if a shifter $\pi^n - \pi^*$ was added to the Phillips curve:

$$\pi = (\pi^n - \pi^*) + \pi^* + \frac{y(\theta)}{\varpi\delta(\delta - r)} \cdot \left[\frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right]. \tag{14.24}$$

Because of the shifter $\pi^n - \pi^* \neq 0$, when the economy is at full employment (and the term in square brackets is 0), inflation is at π^n instead of π^*. So the divine coincidence breaks down: being at full employment does not guarantee price stability.

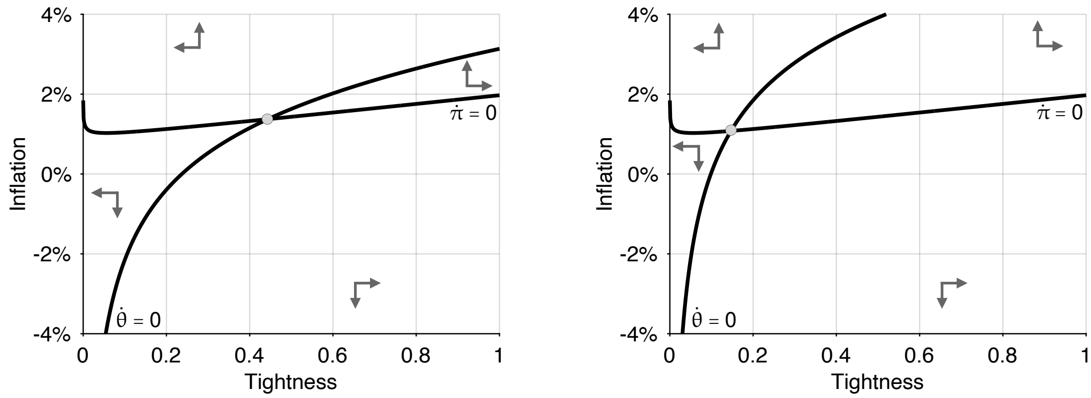
The two mandates of the central bank are not aligned here, and cannot be achieved at the same time. For instance if people believe that inflation will normally be above target ($\pi^n > \pi^*$), then inflation is too high when the economy is at full employment. Equivalently, when inflation is on target, tightness is too low, and the unemployment rate is inefficiently high.

Hence, in this model, the coincidence is not so much "divine" as the result of careful and effective messaging by the central bank.

14.12. Solution of the model at the zero lower bound

In normal times, the central bank sets a real interest rate, which separates the dynamics of inflation and tightness. Tightness can therefore be determined as in the fixed-inflation model of chapter 13, and inflation can be determined from the Phillips curve. At the zero lower bound, this separation disappears as the central bank loses control of the real interest rate. The nominal interest rate is stuck at 0 so the real interest rate is just negative inflation: $r = i - \pi = -\pi$. Tightness and inflation are therefore jointly determined by a two-dimensional dynamical system.

What might cause the economy to reach the zero lower bound? Typically, a large drop in aggregate demand, for instance a large increase in the marginal utility of wealth $1/a$. Aggregate demand must be depressed enough that a positive nominal interest rate is insufficient to maintain the economy at the divine equilibrium, where $\theta = \theta^*$ and $\pi = \pi^*$. The economy enters the zero lower bound only when at a zero nominal interest rate, aggregate demand is below the efficient level of output: $y^d(\theta^*, -\pi^*) < y^s(\theta^*)$. Hence, the marginal utility of wealth is high enough to push the economy to the zero lower bound



A. Sink at the zero lower bound

B. Sink with lower aggregate demand

FIGURE 14.2. Phase diagrams at the zero lower bound in the calibrated model

The $\dot{\pi} = 0$ nullcline is given by equation (I.10). The $\dot{\theta} = 0$ nullcline is given by equation (I.12). The model is calibrated as in table 10.2, with in addition labor productivity normalized to $k = 1$, the discount rate set to $\delta = 91\%$, the real interest rate set to $r = 0.5\%$, and the price-adjustment costs set to $\varpi = 58$. In panel A, the taste for consumption is $a = 1.66$. In panel B, the taste for consumption is lower, set to $a = 1.62$.

when

$$\frac{1}{a} > \frac{(\delta + \pi^*)(1 - \alpha)(c^*)^{-\alpha}}{[1 + \tau^*]},$$

where c^* and τ^* are the efficient values of consumption and the matching wedge.

The tightness-inflation dynamical system describing the model at the zero lower bound is derived and analyzed in appendix I. We report the main results here. The system comprises two differential equations:

$$\frac{\dot{\theta}}{\theta} \cdot [\alpha [1 - \eta(\theta)]\, u(\theta) + (1 - \alpha)\eta(\theta)\tau(\theta)] = \frac{1}{a(1 - \alpha)} \cdot [1 + \tau(\theta)]\, c(\theta)^{\alpha} - (\delta + \pi) \tag{14.25}$$

$$\dot{\pi} = \delta\left[\pi - \pi^*\right] + \frac{a(1 - \alpha)}{\varpi} \cdot c(\theta)^{1-\alpha}\left[1 - \frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}\right]. \tag{14.26}$$

The $\dot{\theta}$ equation arises from equation (14.17), which describes households' consumption and saving behavior. The $\dot{\pi}$ equation arises from equation (14.18), which describes households' pricing behavior.

We construct the phase diagram of the nonlinear dynamical system to illustrate its properties, following the procedure presented in appendix A.8.2. The phase diagram is displayed in figure 14.2, under two possible calibrations.

The $\dot{\theta} = 0$ nullcline is a combination of the aggregate demand and aggregate supply curves from figure 13.1B. As inflation rises, the real interest rate falls, so the aggregate demand curve moves up along the aggregate supply curve, leading to a higher tightness.

The nullcline admits the following expression:

$$\pi = \frac{1}{a(1-\alpha)} \cdot [1+\tau(\theta)]\, c(\theta)^{\alpha} - \delta. \tag{14.27}$$

The $\dot{\pi} = 0$ nullcline is mostly based on the Phillips curve from figure 14.1A, and it mostly describes inflation as an increasing function of tightness. The nullcline is given by

$$\pi = \pi^* + \frac{a(1-\alpha)}{\delta\varpi} \cdot c(\theta)^{1-\alpha} \left[\frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1 \right]. \tag{14.28}$$

Critically, the nullcline remains below the inflation target π^* on $(0, \theta^*)$, and is closer to it when prices are more rigid (higher ϖ).[3]

The system always admits an equilibrium, and when price rigidity ϖ is large enough, the equilibrium is guaranteed to be unique—as in figure 14.2. The large price rigidity keeps the nullcline $\dot{\pi} = 0$ flat enough, so close enough to $\pi = \pi^*$.

In that case, as we can see in figure 14.2, the dynamical system is a source around the equilibrium. Indeed, in all four quadrants of the phase diagram, the trajectories move away from the equilibrium.

This implies that the solution of the model is unique at the zero lower bound—although monetary policy is passive. The only solution is for tightness and inflation to jump to the equilibrium and stay there. If tightness and inflation jumped somewhere else, they would diverge away from the equilibrium.

An advantage of the slackish model is that we can use it to study permanent ZLB episodes. Unlike in the New Keynesian model, indeterminacy is never a risk at the zero lower bound. As we had already seen in chapter 13, the central bank can follow an interest-rate peg without creating indeterminacy. This is true even in this variant of the model with endogenous inflation.

So we can just assume that aggregate demand is depressed and the central bank keeps the interest rate at zero forever. The model admits a unique solution, where the economy is in a slump: inflation is below target and tightness is below its efficient level. The solution at the ZLB is the intersection of the two nullclines: (14.27) and (14.28).

When an unexpected and permanent shock occurs, the economy jumps to a new ZLB solution. We can use the phase diagram to study such jumps. For instance, negative aggregate-demand shocks continue to operate in a similar fashion at the zero lower bound. A negative aggregate demand shock—such as a decrease in the taste for consumption a—leads to an inward shift of the $\dot{\theta} = 0$ nullcline (from figure 14.2B to figure 14.2A). In response to the negative shock, tightness is lower and the unemployment rate is higher. If

[3]Recall that when tightness is inefficiently low, $\theta < \theta^*$, then $\tau(\theta)/u(\theta) < [1-\eta(\theta)]/\eta(\theta)$, as equation (9.11) implies.

the solution is on the upward-sloping part of the $\dot{\pi} = 0$ nullcline, then inflation is lower.

14.13. Empirical evidence on the Phillips curve

In this section, we review recent US evidence supporting the slackish Phillips curve. In the modern US economy, the divine coincidence appears to hold, the inflation gap moves negatively with the unemployment gap, and the Phillips curve in the unemployment-inflation plane appears much steeper below the FERU than above it.

14.13.1. Divine coincidence

Figure 14.3A illustrates the parallel trajectories followed by the inflation gap and unemployment gap in the United States during the pandemic and after (2020–2024). Inflation is measured as annual change in the price index for personal consumption expenditures, which is itself produced by the BEA (2026). The inflation gap is the inflation rate minus an inflation target of 2%. These are the inflation measure and inflation target used by the Federal Open Market Committee (FOMC 2026). The unemployment gap is computed in chapter 12 (figure 12.3).[4]

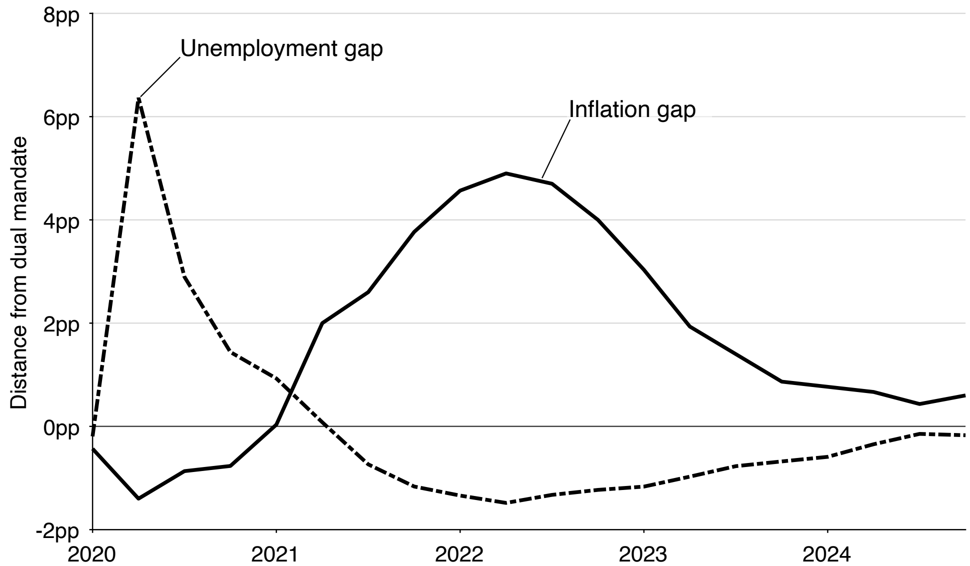
The main observation is that unemployment gap and inflation gap evolve in tandem between 2020 and 2024. Both gaps are close to 0 at the beginning of 2020: the unemployment gap is just −0.2pp while the inflation gap is −0.4pp. So the economy is close to full employment and price stability—close to the point of divine coincidence.

Then, the following quarter, the unemployment gap rises very sharply and the inflation gap drops very sharply. In 2020:Q2, the unemployment gap reaches +6.4pp while the inflation gap bottoms at −1.4pp.

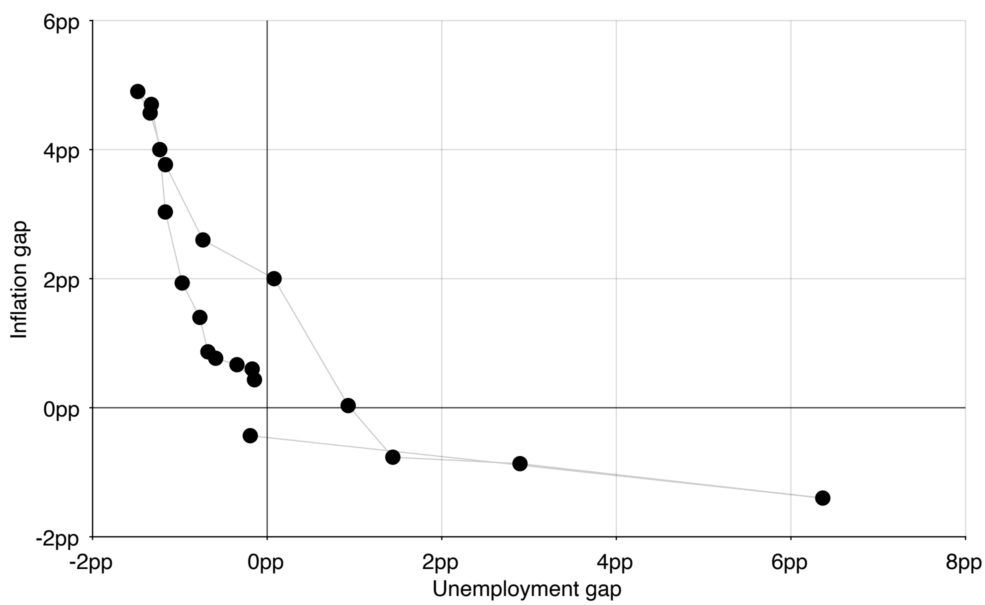
After that both gaps recover. They first return to 0 in 2021:Q1 for inflation and the following quarter for unemployment. Then, the inflation gap keeps rising and the unemployment gap keeps falling. In 2022:Q2, the inflation gap peaks at +4.9pp while the unemployment gap bottoms at −1.5pp.

After reaching these extreme values, both gaps normalize until the end of 2024. The inflation gap shrinks to reach +0.4pp in 2024:Q3. The unemployment gap shrinks to reach −0.1pp in 2024:Q3. So the economy is essentially back at full employment and stable prices at the end of 2024.

[4] The unemployment gap was computed by focusing on efficiency of the labor market, not the entire economy. This restriction is visible in two ways. The unemployment rate was measured from household data (CPS), and therefore does not include the sort of idle labor in firms or unsold goods described in chapter 2, which would ideally be included to compute aggregate tightness. The vacancy rate was measured from firm data (JOLTS), and therefore does not include the sort of buying effort by households described in chapter 2, which should also be included to compute aggregate tightness. Nevertheless, in the absence of better data, we use that labor-market unemployment gap instead of an aggregate slack gap for our analysis.



A. Time series



B. Scatter plot

FIGURE 14.3. Phillips curve in the United States, 2020–2024

The unemployment gap comes from figure 12.3. The inflation rate is the quarterly average of the percentage change from a year ago of the monthly seasonally adjusted personal consumption expenditure price index, which is produced by the BEA (2026). The inflation gap is the inflation rate minus the inflation target of 2%.

This parallel pattern suggests that the divine coincidence prevails in the modern US economy. The inflation gap is 0 whenever the unemployment gap is 0; or the inflation rate is on target whenever the unemployment rate is at the FERU; or prices are stable whenever the economy is at full employment. This coincidence occurred at the beginning of 2020, the beginning of 2021, and the end of 2024.

Moreover, inflation gap and unemployment gap are negatively related: inflation rises above target whenever the unemployment rate is below the FERU; and inflation falls below target whenever the unemployment rate is above the FERU.

The divine coincidence and negative relationship is also clearly visible when the inflation and unemployment gaps are displayed in a scatter plot (figure 14.3B). Interestingly, the empirical Phillips curve drawn by the US economy after the pandemic is quite close to the Phillips curve simulated from the slackish business cycle model (figure 14.1B). For the same fluctuations in unemployment gap, the inflation gap varies by comparable amounts in the data and simulations.

The basic evidence of divine coincidence presented in figure 14.3 is supported by other work. Benigno and Eggertsson (2023) use aggregate data for inflation and labor market tightness in the United States. They find clear evidence of divine coincidence in the 2008–2022 period. When the labor market is efficient, which corresponds to a tightness of 1 (figure 12.1), inflation is on target at 2% (Benigno and Eggertsson 2023, figure 4).

In the 1960s, the Council of Economic Advisors believed in the divine coincidence since they thought that at full employment the economy was not subject to inflationary pressures (Bernanke 2022, p. 18). And over the years the Fed through its FOMC statements often indicated that it believed in the divine coincidence, because it believed that price stability necessarily guaranteed full employment, so that its two mandates were equivalent (Thornton 2012, pp. 119–121, p. 130).

14.13.2. Convexity

Finally, it seems not only that the divine coincidence holds in the United States, but also that the unemployment-inflation Phillips curve appears convex.

The convexity of the Phillips curve can be seen clearly in figure 14.3B. In early 2020, the drop in inflation gap (down to −1.4pp) is much more subdued than the coinciding spike in unemployment gap (up to +6.4pp). So the economy was inefficiently slack and most of the movement occurred with the unemployment gap. By contrast, when the economy became inefficiently tight after 2021, most of the movement occurred with the inflation gap. In 2022, the spike in inflation gap (up to +4.9pp) was much larger than the drop in unemployment gap (down to −1.5pp). This indicates that the unemployment-inflation Phillips curve is much flatter when the economy is inefficiently slack than when it is inefficiently tight.

More generally, Babb and Detmeister (2017, table 4) and Smith, Timmermann, and Wright (2025, table 5) find that inflation is much more responsive to unemployment along the US Phillips curve when the unemployment rate is below 4.2% than when the unemployment rate is above 4.2%. In the postwar United States, the efficient unemployment rate is just 4.2% (figure 12.2), so the kink is detected at the efficient unemployment rate. It is true that these studies describe the pattern as a kink, whereas the slackish Phillips curve is smooth, but a convex Phillips curve can look kinked in finite samples.[5]

14.14. The economy after the coronavirus pandemic

In this final section, we use the slackish business cycle model and its Phillips curve to revisit economic fluctuations after the coronavirus pandemic.

14.14.1. Post-pandemic shocks

We consider next an unusual business cycle shock: a shock that shifts the Beveridge curve and then the aggregate supply curve. Such shock involves a change in either the job-separation rate λ or the matching efficacy μ. Both an increase in job separations and a decrease in matching efficacy shift the Beveridge curve outward, which raises the efficient unemployment rate u^*.

A negative supply shock leads to an inward shift of the aggregate supply curve (figure 13.2B). In response to the negative shock, tightness is higher, but the unemployment rate is higher, and inflation is higher. But the key is that the unemployment gap is lower (it has become negative) and inflation is higher. Indeed the efficient unemployment rate has increased more than actual unemployment, so the unemployment rate is now inefficiently low. Such excessive tightness leads to higher inflation.

If the Euler curve remains the same after the outward shift of the Phillips curve—for example because the central bank is not aware of it—then we obtain a burst of inflation after the adverse shock to the Phillips curve. Since the US Beveridge curve has shifted dramatically outward in the aftermath of the coronavirus pandemic (figure 2.7), the flare-up in inflation in 2021–2023 might partly result from this dramatic shift.

The model predicts that inflation rises above target whenever the labor market is inefficiently tight. The model therefore provides an explanation for the flare-up in inflation

[5] Benigno and Eggertsson (2023, figure 4) even find that inflation is more responsive to labor market tightness along the US Phillips curve when tightness is inefficiently high (above 1) than when tightness is inefficiently low (below 1). In the numerical illustration of figure 14.1A, the slackish Phillips curve does not display this property: it is essentially linear in tightness. One way to generate the nonlinearity observed by Benigno and Eggertsson is to introduce an asymmetric price-adjustment cost, as Michaillat and Saez (2024) do: when wage cuts are more costly than price hikes, the tightness-inflation Phillips curve is steeper when the economy is inefficiently tight than when it is inefficiently slack.

in 2021–2023. The US labor market was inefficiently tight after the coronavirus pandemic—in fact tighter than at any point since the end of World War 2 (figure 12.1). That excessive tightness must have fueled the flare-up in inflation, all the more so because the Phillips curve $\pi(u)$ is convex: a given tightening of the labor market moves inflation more when unemployment is already low than when unemployment is already high.

14.14.2. Soft landing

After the pandemic, inflation was too high: the inflation gap peaked at 5pp in 2022 (figure 14.3). The Fed wanted to achieve a soft landing: to reduce inflation without raising unemployment.

However, in our model, a strict soft landing was not desirable, since unemployment was too low: the unemployment gap bottomed at -1.5pp in 2022. Moreover, in the model, a soft landing is not possible.

Monetary policy operates through the aggregate demand curve. To reduce inflation, tightness must fall, so monetary policy must raise the real interest rate (figure 13.2A). From this, we can see that strict soft landing is not possible—the unemployment rate always increases after a decrease in tightness.

Even though a strict soft landing isn't possible, unemployment might not increase by much, especially when it is low. This is because the aggregate supply curve is convex in slackish models, as we established in chapter 8. When tightness is high and unemployment is low, the aggregate supply curve is steep so as tightness is reduced, and tightness falls, the unemployment rate doesn't increase by a lot. Thus, it is true that if the central bank cools the economy at a spot where there is little unemployment, a reduction in tightness generates a small increase in unemployment. On the other hand, if we try to cool the economy in the flatter region of the aggregate supply curve, a further reduction in tightness increases the unemployment rate a lot. Thus, a "loose" soft landing is possible with our convex aggregate supply curve: the increase in unemployment might be small.

In our model, we can compute the elasticity of the unemployment rate to tightness along the aggregate supply curve—and relate it to the elasticity of the Beveridge curve, which we have measured (chapter 2). By definition, $\theta = v(u)/u$, where $v(u)$ is the Beveridge curve—which is isomorphic to the aggregate supply curve. Implicitly differentiating in elasticity, we get

$$1 = -\beta \cdot \epsilon_{\theta}^{u} - \epsilon_{\theta}^{u},$$

where $\beta = -\epsilon_{u}^{v}$ is the elasticity of the Beveridge curve. Hence, the elasticity of the unemployment rate to tightness is directly determined by the Beveridge elasticity:

$$\epsilon_{\theta}^{u} = \frac{-1}{1+\beta}.$$

In the United States, $\beta = 1$, so $\epsilon^{u}_{\theta} = -1/2$.

Therefore, if the Fed wants to bring labor market tightness down from 2, its value at the peak of the post-pandemic boom (2022:Q2) to 1 (the pre-pandemic, efficient level), the unemployment rate would increase by $50\%/2 = 25\%$, since labor market tightness falls by 50%. Given that the unemployment rate was 3.5% at the peak of the post-pandemic boom (2022:Q3), it would increase to $3.5\% \times 1.25 = 4.4\%$. Thus, the effect on unemployment wouldn't be extreme, but it would not be zero either. In fact, the increase predicted by the Beveridge curve matches almost exactly the increase in unemployment observed after the pandemic: in July 2025, labor market tightness reached 0.99, and the unemployment rate was at 4.3% (Michaillat and Saez 2026).

14.15. Summary

This chapter develops a slackish model of the Phillips curve that links inflation to economic slack. Price dynamics are driven by the amount of slack in the economy. When the economy is inefficiently slack, sellers reduce their prices to attract customers. Conversely, when the economy is inefficiently tight, sellers raise their prices. Without price rigidity, the model would be efficient, as in Moen (1997)'s model. But with price rigidity, the model generates a Phillips curve that describes inflation as a function of the unemployment gap.

The slackish Phillips curve ensures divine coincidence: inflation is on target if and only if the unemployment rate is efficient. In the unemployment-inflation plane, the Phillips curve is strictly convex in unemployment (see also appendix H). Inflation is therefore more sensitive to the unemployment gap when the labor market is inefficiently tight than when it is inefficiently slack.

Recent US evidence supports the model's predictions. The divine coincidence holds: inflation rose above target in 2021 as the unemployment rate fell below the FERU, and inflation has declined since its 2022 peak as unemployment has moved back toward the FERU. The unemployment-inflation Phillips curve also appears steeper when the labor market is inefficiently tight than when it is inefficiently slack.

Bibliography

Babb, Nathan R. and Alan K. Detmeister. 2017. "Nonlinearities in the Phillips Curve for the United States: Evidence Using Metropolitan Data." Finance and Economics Discussion Series 2017-070. https://doi.org/10.17016/FEDS.2017.070.

BEA. 2026. "Personal Consumption Expenditures: Chain-type Price Index." FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/PCEPI.

Benigno, Pierpaolo and Gauti B. Eggertsson. 2023. "It's Baaack: The Surge in Inflation in the 2020s and the Return of the Non-Linear Phillips Curve." NBER Working Paper 31197. https://doi.org/10.3386/w31197.

Bernanke, Ben S. 2022. *21st Century Monetary Policy*. New York: W. W. Norton.

Cochrane, John H. 2011. "Determinacy and Identification with Taylor Rules." *Journal of Political Economy* 119 (3): 565–615. https://doi.org/10.1086/660817.

FOMC. 2026. "Statement on Longer-Run Goals and Monetary Policy Strategy." https://perma.cc/7BBR-9UFG.

Hazell, Jonathon, Juan Herreno, Emi Nakamura, and Jon Steinsson. 2022. "The Slope of the Phillips Curve: Evidence from U.S. States." *Quarterly Journal of Economics* 137 (3): 1299–1344. https://doi.org/10.1093/qje/qjac010.

Michaillat, Pascal and Emmanuel Saez. 2024. "Beveridgean Phillips Curve." arXiv:2401.12475. https://doi.org/10.48550/arXiv.2401.12475.

Michaillat, Pascal and Emmanuel Saez. 2026. "Automated Business Cycle Dashboard." https://pascalmichaillat.org/dashboard/.

Moen, Espen R. 1997. "Competitive Search Equilibrium." *Journal of Political Economy* 105 (2): 385–411. https://doi.org/10.1086/262077.

Okun, Arthur M. 1981. *Prices and Quantities: A Macroeconomic Analysis*. Washington, DC: Brookings Institution.

Phillips, A. W. 1958. "The Relation Between Unemployment and the Rate of Change of Money Wage Rates in the United Kingdom, 1861–1957." *Economica* 25 (100): 283–299. https://doi.org/10.1111/j.1468-0335.1958.tb00003.x.

Rotemberg, Julio J. 1982. "Monopolistic Price Adjustment and Aggregate Output." *Review of Economic Studies* 49 (4): 517–531. https://doi.org/10.2307/2297284.

Smith, Simon, Allan Timmermann, and Jonathan H. Wright. 2025. "Breaks in the Phillips Curve:

Evidence from Panel Data." *Journal of Applied Econometrics* 40 (2): 131–148. https://doi.org/10.1002/jae.3102.

Thornton, Daniel L. 2012. "The Dual Mandate: Has the Fed Changed Its Objective?" *Federal Reserve Bank of St. Louis Review* 94 (2): 117–134. https://doi.org/10.20955/r.94.117-134.

Zbaracki, Mark J., Mark Ritson, Daniel Levy, Shantanu Dutta, and Mark Bergen. 2004. "Managerial and Customer Costs of Price Adjustment: Direct Evidence from Industrial Markets." *Review of Economics and Statistics* 86 (2): 514–533. https://doi.org/10.1162/003465304323031085.